

ORIGINAL RESEARCH

A blockchain-based secure framework for data management

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Abstract

Data management is a crucial requirement due to the autonomous and constrained nature of Unmanned Aerial Vehicles (UAVs), Internet of Things (IoTs), and the aviation domain. The autonomous and restricted nature of these sectors increases the need for a shared, distributed database, strong access control management, consensus in autonomous decision-making, and effective communication across diverse protocols and devices. This research presents a comprehensive approach and offers a new viewpoint to the field of blockchain while establishing a fundamental baseline for future improvements in data management systems and addressing the shortcomings of previously proposed existing frameworks in order to fulfill the complex needs of secure data management. This study contributes to the advancement of secure and efficient data management systems by implementing robust data monitoring for error detection, ensuring data integrity, and enabling encrypted or anonymous data sharing based on sensitivity levels. Additionally, the integration of diverse devices, enforcement of immutable regulations compliance, and development of permissioned blockchain systems for identity management further enhance the system's capabilities, offering comprehensive solutions for modern data management challenges. In the tests, the proposed framework showed increased successful transactions in all rate controllers. Besides, effect of the validator number on throughput and latency is tested and analyzed thoroughly.

1 | INTRODUCTION

The functionalities of blockchain technology are continuously expanding due to its inherent decentralization, lack of reliance on trust, unchangeable storage, and ability to share information anonymously. Therefore, the use of blockchain-based applications in diverse sectors is consistently growing: Insurance [1], health [2], land registry cadastre [3], voting [4], industry [5], finance [6], supply chain [7], social media [8], and ticket sales [9] are among the widespread utilization areas. There is a growing interest in the utilization of blockchain technology in the aviation industry, the Internet of Things (IoT), and the area of Unmanned Aerial Vehicles (UAV) [10]. The independent and limited nature of these areas highlights the crucial significance of maintaining data accuracy and ensuring the reliability of its sources. In the face of these difficulties, blockchain technology arises as a viable remedy, praised for its ability to enhance data security through features such as immutability, authentication, authorization, and encryption. The application

of this technology is wide-ranging, encompassing smart cities, IoT, aviation, and UAVs. Its primary goal is to reduce privacy breaches and safeguard personal data [11, 12]. The growing popularity of blockchain technology can be attributed to its distinct advantages, such as enhanced security, decentralization, and fault tolerance. These benefits are anticipated to entirely revolutionize existing operational systems.

Entities in the aviation, UAV, and IoT industries collaborate in clusters at times and operate independently at other times, as seen in Figure 1. Attributes such as data integrity, confidentiality, availability, increased access control, traceability, and seamless integration are commonly recognized as vital in various sectors. The inherent characteristics of blockchain effectively address these challenges. As exemplified, Umran et al. have utilized the characteristics of blockchain to address the privacy and security needs related to data exchange in the field of Industrial Internet of Things (IIoTs) [13]. Similarly, Raj et al. have suggested a customized access control system specifically designed for monitoring systems in the Healthcare

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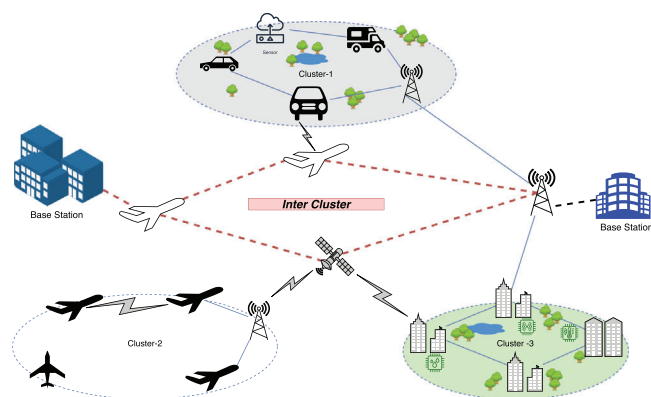


FIGURE 1 An overview of the problem domain.

domain [14]. This validates the effectiveness of the technology and its appropriateness for these fields. Nevertheless, there is a notable deficiency in a comprehensive framework that effectively handles the intricate requirements of safe data management in these crucial fields, despite the acknowledged promise of blockchain technology [15] and [16].

The growing scholarly focus on blockchain not only emphasizes its significance within specialist domains but also indicates a broader transition toward a secure and interconnected digital future. The incorporation of blockchain technology in many industries exemplifies its pivotal function in fostering the creation of safe, streamlined, and user-centric digital frameworks. Furthermore, the literature analysis provides insight into particular blockchain applications and the difficulties encountered in these areas. To assess the influence of blockchain applications on COVID-19, a bibliometric study is conducted to identify the latest areas of research [17]. The study identified three primary areas of inquiry: the utilization of blockchain technology in organizational settings, specifically focusing on research methodologies in artificial intelligence; the imperative to tackle business sustainability issues stemming from the COVID-19 pandemic; and the ramifications of the pandemic on the supply chain industry. Further research is dedicated to exploring blockchain ideas in the domains of intelligent transportation and aviation with the aim of improving privacy protection and data management [18]. Although the smart cities community is mainly concerned with protecting personal data that is processed, exchanged, or stored, there is still a worry about maintaining privacy in intelligent transportation systems that use Unmanned Aerial Vehicles (UAVs) in the context of smart cities [12]. In the aviation industry, flight data recorders are being used on the blockchain to replay flight events to comply with the Flight Operations Quality Assurance (FOQA) criteria [19, 20]. Another instance in the aviation industry has utilized blockchain technology to handle and safeguard flight data, showcasing its effectiveness in improving operational efficiency [21]. The cross-sectoral interaction and integration of blockchain applications demonstrate the versatility and potential of the technology as a crucial element of future digital systems. An additional instance of cross-sector collaboration and integration of blockchain applications is discussed in [22].

This study showcases the utilization of blockchain technology to enhance security in the domain of Industrial Internet of Things (IIoT) for Electric Smart Grid. Despite the existence of successful pilot projects and theoretical endorsement of the value of blockchain, it is crucial to acknowledge that the technology is still in its nascent phase in terms of practical application in industries and commerce [23]. However, blockchain is making significant advancements to achieve the necessary level of development for its widespread implementation in industrial and commercial environments.

After investigating the primary issues, as examined and outlined in Section 2, found that these domains are found to encompass inadequate storage capacity, unreliable communication links, insecure data-sharing channels that prioritize data confidentiality, substantial administrative requirements for maintaining connectivity, lack of transparency in central authority management for users, diverse devices and communication protocols, vulnerability to single points of failure, absence of self-management capabilities, insufficient identity management in a distributed environment, and non-scalable frameworks for expansion. Although there have been significant improvements in blockchain applications, there is still a need for a comprehensive framework that encompasses the fundamental principles of blockchain in order to effectively handle the overall difficulties of data management. This study seeks to address this gap by presenting a framework that not only guarantees security and confidentiality, but also facilitates smooth communication across various protocols, upholds self-management in accordance with confidentiality requirements, and preserves unchangeable data in a dispersed domain. This environment necessitates the use of a distributed database, efficient data management, cooperation, secure data sharing, decentralized decision-making, and communication across various protocols and devices. Although extensive research has been conducted on data management in these fields, there is a notable absence of a comprehensive framework expressly designed to address these data management challenges.

To solve the aforementioned challenges and make a great start for further improvements, this study introduces a thorough and flexible framework for guaranteeing the secure administration of data. It employs blockchain technology in the aviation, Internet of Things (IIoT), and unmanned systems sectors. The framework stands out due to its unique clustered architecture, intentionally created to address deficiencies in current research, enhance scalability, and efficiently manage the complexities of node interactions in extensive networks. This architectural design allows for the integration of tailored consensus mechanisms that align with the specific operational needs of the framework. The system employs a stringent methodology to safeguard data from both external and internal threats. The framework efficiently fulfills the security requirements for managing sensitive data by implementing role-based access control, as well as authentication and authorization techniques. This emphasizes the framework's commitment to providing absolute security.

Moreover, the model's hierarchical framework enables smooth integration and ongoing development, showcasing its

capacity to foresee technological improvements and the necessity for digital adaptability. The classification of segments into inter-cluster and intra-cluster categories, together with a modular organization of contracts and duties, ensures the efficient and secure storage and retrieval of data. Moreover, this feature provides the adaptability essential to meet the specific demands of various industries. The suggested framework aims to address present and future data management challenges in different industries by employing a clustered approach, stringent security measures, and a customizable design, thereby offering a robust and tailored solution. Integrating this intricate framework into modern systems is anticipated to make a significant contribution to the advancement of autonomous, secure, and efficient digital infrastructures. Our main contributions to this study encompass:

- Enabling the secure exchange of information among nodes through cryptographic methods involves allowing data sharing in either encrypted or anonymous formats, depending on the sensitivity level of the material. This method involves protecting the data's integrity, which means making sure it remains unchanged, intact, and free from any illegal interventions or modifications.
- Integrating a diverse array of devices including drones, IoT devices, and airplanes, each designed to carry out certain tasks, while differing in their processing abilities and access to blockchain technology. This entails enforcing compliance with unchangeable regulations by all participants throughout the whole process.
- Efficiently updating databases is accomplished by utilizing the structure of the distributed framework and minimizing storage requirements through the use of inter-cluster and intra-cluster designs. This involves facilitating smooth communication and exchange of information within and between clusters, possibly integrating different blockchain technology.
- Creating a customized blockchain system with restricted access, specifically designed for managing identities. This system offers limited visibility to ensure the confidentiality of personal or confidential corporate information. This entails implementing authentication and authorization systems to meet access management requirements. Authorized individuals are given special permission to carry out data inquiries on the blockchain. They can utilize monitoring tools to analyze operations and ensure compliance with regulations. In addition, incorporating data monitoring capabilities facilitates the detection of errors, allowing for analysis in both online and offline situations.
- The proposed framework demonstrated a higher number of successful transactions across all rate controllers. The impact of the validator number on throughput and delay is investigated.

The subsequent sections of this study are outlined as follows: Section 2 provides the background in the field and our motivation, focusing on the difficulties and proposed solutions discussed in the existing literature. Next, a thorough examination of the literature is conducted to identify similar

studies, followed by a full comparison and analysis. Next, the system concept of the secure management framework based on blockchain is thoroughly explained in the themes depicted in Figure 2. Ultimately, our proposed framework is implemented, an experimental analysis is carried out and our obtained results are presented in order to verify its suitability, usability, and relevance.

In order to evaluate our proposed framework, an Ethereum platform is employed to construct a consortium network of blockchains, where experiments are conducted to initialize data on the chain. The evaluation bed is initialized and tests are run using our developed application that utilizes Hyperledger Caliper to benchmark the framework in terms of throughput and latency. As stated in Section 7, the findings demonstrate that the suggested framework is applicable, suitable, acceptable, scalable, and viable. However, additional examination and advancement are scheduled to guarantee the level of readiness as we prepare for upcoming tasks.

2 | BACKGROUND AND MOTIVATION

Blockchain offers decentralized governance and removes the need for trust in centralized authorities by proposing consensus mechanisms [24]. Consensus is to reach an agreement on the issue without selecting a leader and to distribute rights. Bitcoin proposes the first blockchain solution to the problem and is marketed as a viable financial alternative to traditional currencies by mitigating the need for a central bank [25].

Blockchain is defined as a series of blocks that are linked together in a sequential chain by combining new blocks that contain a header, timestamp, the cryptographic summary of the previous block, and a body that stores the information. The fundamental component of blockchain technology is a series of unchangeable data record blocks that may be confirmed by cryptographic procedures and techniques, including Merkle tree and hash algorithms. Currently, some studies have proposed replacing the traditional Merkle tree approach with an efficient verification mechanism called the incremental aggregator sub-vector commitment (IASVC) [26]. This change aims to reduce communication usage, overhead, and time by eliminating the need for a Merkle tree and effectively reducing the size of the evidence. It is commonly acknowledged that the immutability of blockchain is based on its strong cryptographic capabilities, regardless of any alterations in the method used. This is accomplished by establishing a connection between each block and the one that comes before it. Therefore, the blockchain can be described as an immutable collection of records with no need for trust in the environment [27].

Reaching a consensus without central authority between any involved party has become available with the help of the blockchain [28]. The most widely accepted and used solutions depend on blockchain technology [29].

When the benefits came to the fore after releasing Bitcoin, different blockchain technologies emerged with diverse capabilities [30]. The leading innovation that diversifies the usage domain of blockchain is the smart contracts which is

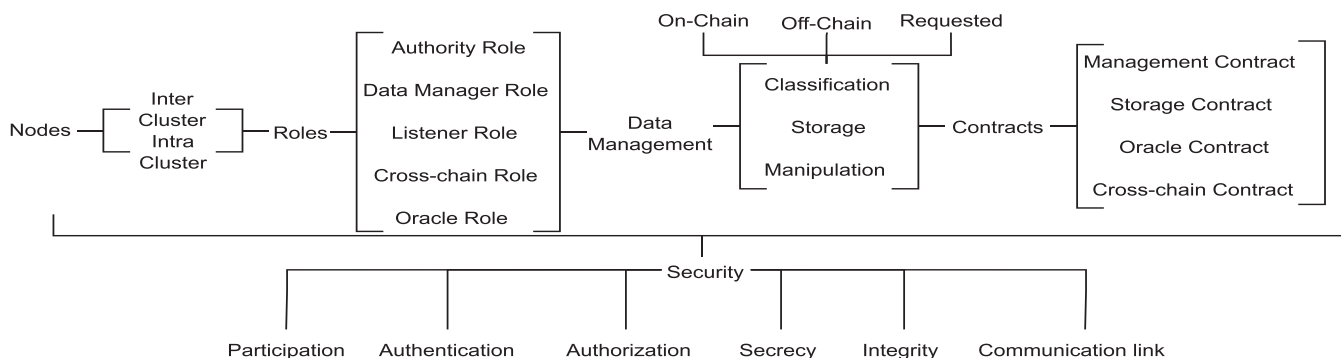


FIGURE 2 The system model of the proposed blockchain-based secure management framework.

proposed by Ethereum [31]. While Bitcoin technology is only capable of creating unspent transaction output (UTXO) to store monetary transactions, Ethereum blockchain is a Turing complete, programmable, solution that enables building distributed trusted applications. In other words, blockchain technology has advanced from being an unchangeable cryptocurrency trade ledger into a programmable interactive environment [32]. Another initiative, Hyperledger, develops a business consortium blockchain to address all types of requirements for an open-source, cryptocurrency-supported, adaptable, and secure blockchain [33].

Conventional information technology systems, social media platforms, and other services that depend on users' data necessitate obtaining consent before utilizing the data, and it is imperative to maintain the confidentiality of this information. Nevertheless, as demonstrated by the Wikileaks incident, the act of revealing sensitive and classified information can result in detrimental consequences for the economy, politics, and human rights [34]. Facebook's influence was evident in both the 2016 US presidential election and the Taiwanese presidential election, as highlighted by several sources [35, 36]. These occurrences highlighted the need to treat every piece of data as valuable and refrain from sharing it without the owner's consent. Information has become a potent force and a valuable resource on a global scale. Blockchain is a potential remedy for these problems, as it is known for safeguarding the confidentiality of personal information by employing cryptographic identification and access control methods that ensure anonymity.

Blockchain is extensively employed across various domains, including smart cities, IoT, aviation, and UAV, to address the issues arising from privacy breaches. These domains are inter-related and share common objectives within their respective fields. The smart cities community is primarily focused on safeguarding the personal data that is processed, exchanged, or stored [11]. Additionally, the use of Internet of Things (IoT) devices has opened up new research opportunities in the field of smart cities, which are considered to be complex and demanding. Ensuring data confidentiality is of utmost importance since individuals with access to personal navigation or location monitoring data can disclose information about their

habits, acquaintances, employment, preferred places to visit, interests, and other potential issues.

Hirtan et al. [18] describe a blockchain-based architecture for privacy protection in the intelligent transportation industry, which is related to the previously stated areas. The proposed design consists of a primary data server and a processing server to cater to the requirements of smartphones' efficient communication and processing capabilities. Upon receiving the enrollment phase shared key, the server obtains knowledge of the secret key. Consequently, the central authority changes, while it remains in existence. To ensure identity anonymity, it is advisable to deploy the architecture in a trusted environment or with a permissioned blockchain.

In the aviation industry, blockchain technology is primarily utilized to address several challenges related to data management, sharing, storage, and analysis. A crucial component in aviation known as a "Flight Data Recorder" (FDR) is designed to remain operational even in the event of an accident [21]. The responsibilities of this system are monitoring aircraft, enhancing safety measures, finding issues, and predicting probable future accidents. The primary characteristic of frequently utilized Flight Data Recorders (FDR) is their capacity to retain data in an immutable and easily accessible manner, while also being able to endure impact shock, penetration, static crushing, high-temperature fire, and liquid immersion following an accident [37]. Research has been conducted in the literature to decrease the dimensions and mass of these large and cumbersome containers for Unmanned Aerial Vehicles (UAVs). Digital flight data recorders have evolved to preserve increasingly accurate data and enable the playback of flights to identify any anomalies related to flight safety. The Red Cat company has developed a flight recorder for unmanned aircraft systems (UAS) that utilizes blockchain technology [38]. This recorder allows for the analysis and playback of flight data, which is encrypted for security purposes. Additionally, the recorder is equipped with artificial intelligence (AI) capabilities to assist with diagnostics. Moreover, this paper introduces a distributed storage architecture based on blockchain technology, which ensures the secure transport and storage of data by leveraging encryption techniques. The solution they provide has an open-source and adaptable

structure, aiming to support regulators, insurance companies, and drone fleet operators.

In addition to safeguarding and guaranteeing the accuracy of the data contained in the Flight Data Recorder (FDR) to identify the reasons for a crash or detect any anomalies during the flight through deterministic playback, it is necessary to analyze the flight data. It plays a crucial function in both the promptness of the data that is retrieved and the accuracy of the data that is recorded. The research undertaken to document flight records and capture access status inside the utilized data carefully considers deterministic replay, operating system input and output (OS-I/O) management, and hardware support [20]. The primary objective is to furnish all the necessary aeronautical information for subsequent playback of the flight. Technological improvements have facilitated the conversion of flight data recorders (FDRs) into digital format and the utilization of recorded data for anomaly identification, hence enhancing flight safety in the future. The utilization of recorded onboard data analysis, studies, and diverse anomaly detection procedures is employed to bolster the standards of Flight Operations Quality Assurance (FOQA) [39, 40].

Every day, more and more commercial applications in the aviation industry that focus on utilizing blockchain technology to use correct data are being developed [41, 42]. The potential of blockchain technology to revolutionize data management and enhance security across several industries is increasingly recognized as we progress into the digital age. This recognition is evidenced by a substantial surge in scholarly research aimed at addressing industry-specific challenges through the implementation of blockchain applications, fostering a fundamental shift towards systems that are characterized by enhanced security, transparency, and efficiency. Researchers and corporations are studying the industrial applications of UAVs, drones, and IoTs [43]. Jensen et al. aims to demonstrate the practical applications and technological advancements of blockchain in terms of enhancing security, ensuring availability, and maintaining immutability, going beyond typical use cases and scenarios [44]. The study discusses the possible application of blockchain technology in the stated areas, but it does not include the practical implementation or a comparison of different blockchain systems. Boeing and SparkCognition collaborated to create SkyGrid, a company that offers a complete system for unmanned and autonomous aircraft in the aviation industry [23]. This is another example of how blockchain technology is being used in commercial applications. Once the company is established, it aims to ensure the secure handling of package transfers via drones, as well as the administration of air taxis and other aircraft. The management of UAS traffic is being facilitated by the utilization of blockchain technology, while the requirements of businesses will be addressed through the assistance of artificial intelligence. Additionally, they intend to collaborate with regulatory agencies. Their infrastructure is designed to facilitate both commercial and personal aviation travel.

In addition to the development of commercial applications, academic research is performed to ensure compliance with aviation rules for the use of UAVs, drones, and airplanes. Aviation regulatory authorities prohibit the use of unmanned aerial

vehicles near airports, military areas, and critical infrastructure [45–47]. In addition, the increasing number of affordable and cost-effective drones has made the sky unpredictable, requiring stricter regulations. Clarke et.al. conducted a comprehensive review of the hazards posed by drones to public safety and behavioral privacy to elucidate these concerns [48]. Wild et.al. conducted a comprehensive analysis of 152 incidents involving remotely piloted aircraft systems (RPASs) that occurred between 2006 and 2015. The purpose of this analysis was to identify and assess the potential dangers associated with unmanned vehicles, often known as drones, to develop effective measures for risk prevention [49]. Their research demonstrates that unregulated technical progress in aviation poses a greater risk than pilot mistakes. Regulating these small aerial vehicles is a contemporary issue in aviation safety. There have been a multitude of drone-related incidents in the aviation business. In 2015, a collision between a drone and a plane occurred during a drug smuggling operation in Mexico [50]. In 2017, a drone collided with an aircraft in Canada [51]. Fortunately, the aircraft made a safe landing with minimal damage, although the flight was subsequently canceled. In 2018, Gatwick Airport saw the cancellation or diversion of over 58 flights due to the presence of two drones flying over the airport [52]. In 2019, there was another incident of flight cancellations at Heathrow Airport in England, which was caused by drones [53]. To tackle this problem, Kapitonov et. al. have put up a framework called AIRA, which specifically aims to provide the availability of dependable data for the management of semi-autonomous robots [54]. The susceptibility of the environment in which robots operate and strategize their next action in the cyber-physical realm highlights the importance of reliable data. The Drone Employee project utilizes the AIRA protocol to verify that the UAV adheres to the designated route by accessing data recorded on the blockchain by the dispatcher [55]. The suggested protocol is straightforward, yet it enables the establishment of air traffic management and communication among multiple parties.

Drone or IoT device assaults can happen through two means: the transmission of collected data from the drone to the central system, and the reception of new commands from the control center to the drone. Both types of communication, including data provenance and permission, are susceptible to alterations and negative outcomes [56, 57]. Ensuring the security of the channel is crucial for safeguarding the storage, confidentiality, integrity, and availability of the information. The majority of Internet of Things (IoT) devices and Unmanned Aerial Vehicles (UAVs) are interconnected and rely on cloud servers to meet their computational requirements. This cloud infrastructure is considered secure and helps address the limited storage capacity of small devices [58]. These devices are typically deployed in insecure areas or locations that are susceptible to attacks [59]. Another aspect of importance in the field is the positioning and extent of their coverage, as well as their connectivity [60]. To tackle these issues DroneChain functions as a mediator between the cloud server and the drones to resolve the problem of insecure communication channels [61]. The objective is to guarantee the integrity of the data and commands gathered, making them immune to tampering and available for examination by

external entities. The proposed paper thoroughly analyzes and tests DroneChain to examine its performance to guarantee the integrity of drone data during the transfer and storage of cloud data. Conversely, drones reliably receive commands by storing hash values of directives from the central system or data from other drones. However, because of its reliance on unchangeable data on the blockchain and in the cloud, the design does not offer a complete communication solution.

A research project called FlightChain, presented by SITA, aims to utilize the immutability quality of blockchain to store flight data in the aviation industry [62]. FlightChain focuses on blockchain applications for data management and surpasses the scope of IoT devices and UAVs. Blockchain is often recognized as a decentralized and unchangeable database. The proposed model utilizes a permissioned blockchain architecture, enabling authorized entities to access data. British Airways, Geneva Airport, Heathrow, and Miami International Airport are included in the tests. FlightChain has utilized smart contracts built on Ethereum and Hyperledger Fabric to modify and store over two million flight records. FlightChain's issues persist without resolution or satisfactory answers. The implementation of permissioned blockchain has led to management issues due to the system's lack of self-management capabilities. In reality, it is not a challenge, but a chance to oversee a network that can be expanded and is partially transparent. This network also enables agreement on important management decisions through the use of smart contracts. Hyperledger Fabric is a more advanced platform than Ethereum. However, it has a transaction output of nearly 100,000 transactions per second, which is more than enough. Testing exclusively utilizes limited flying data. The primary objective is to prevent the replication of flight data within a blockchain. The test case focuses on the fundamental capability of the blockchain, which is the simple sharing of data across a network. Therefore, this study is crucial for future research in related fields.

3 | LITERATURE REVIEW

In response to the previous research and difficulties, this study has constructed a framework by taking into account pertinent studies in the literature and resolving any shortcomings or areas that need enhancement. To address this, we undertook a thorough examination of existing literature to evaluate data management systems based on blockchain technology that particularly address difficulties within various domains. The objective was to identify research projects that concentrate on data management, namely those that suggest frameworks or methods to tackle difficulties specific to a particular field. The selection of relevant studies was driven by particular criteria, which included their relevance to the study question and the proposed answer.

In the realm of aviation management, the utilization of blockchain technology is exemplified by the Blockchain-Based Flight Operation Data (BFOD) sharing scheme and the BlockTrust model [63]. These models demonstrate how blockchain can ensure data security and enhance operational efficiency

in the aviation industry [64]. The BFOD system utilizes cryptographic techniques, including zero-knowledge proofs, to enhance the confidentiality of flight operating data. Empirical evidence has demonstrated a significant decrease in the duration required to achieve a consensus. BlockTrust is a notable advancement in the development of reliable aviation information systems. It accomplishes this by creating a secure protocol for delivering data on Air Traffic Management (ATM) networks, notwithstanding potential communication delays caused by the implementation of blockchain technology.

Furthermore, there exists a data-sharing platform that is specifically tailored for aviation vendors [65]. This platform showcases the utilization of blockchain technology to enhance transparency and tackle the persistent issue of restricted information availability in supply chains. These advancements combined demonstrate a favorable trajectory in aviation management towards a cohesive ecosystem enabled by blockchain technology. This indicates a transition from isolated solutions to a comprehensive framework inside the industry.

Similarly, the healthcare sector employs blockchain technology to protect sensitive patient data and empower individuals with greater control over their medical records. HierChain's hierarchical storage solution improves the management of health data across many blockchain platforms, maintaining a balance between the security and performance of IoT networks [66]. Simultaneously, the Health Information Exchange (HIE) is a patient-centered approach for sharing health information. It employs smart contracts to enhance privacy and empower patients to have autonomous control over their health records. The incorporation of blockchain technology, together with decentralized principles, in the healthcare sector emphasizes a forthcoming era where data management is not only exceptionally secure but also prioritizes individuals' privacy. An illustrative instance of this is the study conducted by Nguyen et. al., which introduced a robust intrusion detection system utilizing blockchain technology for secure data transmission [67]. The researchers employed a classification model specifically designed for Cyber-Physical Systems (CPS) in the healthcare sector. Through extensive simulations, they demonstrated that their proposed methodology is remarkably precise, efficient, and practical.

The domain of the Internet of Things (IoT) showcases how blockchain technology optimizes data management and boosts operational efficiency. A decentralized approach to data management is achieved by the combination of clustering, edge computing, and blockchain within a three-tiered architecture. This architecture seeks to tackle the increasing worries over privacy and trust in IoT networks [68]. The MSLShard architecture utilizes a sophisticated adaptive network sharing technique to improve the scalability and security of shared IoT resources. It is crucial for the complex interconnections of smart city infrastructures [69].

The remaining portion offers a comprehensive examination and evaluation of the studies mentioned in the previous section. The assessment of these studies was performed by taking into account attributes such as the "solution domain, layered structure, cluster methodology, storage capacity, access management

TABLE 1 A comparison of similar studies in the literature.

Paper	Solution domain	Layer	Cluster methodology	Storage ability	Access management ability	Test bed applicability	Secrecy-preserving
Proposed Framework	Aviation	Blockchain, data management, access management, link, presentation	Inter-cluster, Intra cluster, hybrid	On-chain, off-chain (proof on-chain), requested data	RBAC	YES	PKI, Offloading data, ACL
[63]	Aviation	N/A	N/A	Off-chain (local, cloud, proof on-chain)	RBAC	N/A	Zero-knowledge proofs
[64]	Aviation	N/A	N/A	On-chain	PKI	N/A	N/A
[70]	Aviation	N/A	N/A	On-chain	PKI	N/A	N/A
[71]	Aviation	N/A	N/A	On-chain	N/A	N/A	N/A
[72]	Healthcare	Application, control, infrastructure	N/A	On-chain	PBAC	YES	PKI
[73]	Healthcare	N/A	N/A	Off-chain (proof on-chain)	ACL	N/A	PKI
[74]	Healthcare	N/A	N/A	Off-chain (cloud, proof on-chain)	ACL	N/A	Zero-knowledge proofs
[75]	Supply chain	Perception, blockchain, application, user	N/A	On-chain	ACL	N/A	PKI
[76]	Supply chain	N/A	Global and local	N/A	ABAC	N/A	N/A
[77]	IoT	N/A	N/A	Off-chain (proof on-chain)	RBAC	N/A	N/A
[78]	Industry	Perception, off-chain, blockchain, application, service	N/A	Off-chain (proof on-chain)	N/A	YES	PKI
[79]	Smart cities	Detection, security, communication, link	N/A	Off-chain(IPFS)	PBAC	N/A	N/A

capabilities, testbed applicability, and preservation of secrecy.” The results of this systematic literature analysis, which provide a systematic comparison of pertinent studies, are organized in Table 1. This table provides a comparative analysis of the proposed framework and other studies, highlighting their level of suitability or lack thereof. When a situation does not apply, the abbreviation “N/A” is utilized. After conducting a thorough analysis, it is evident that blockchain technology continuously supports the goals of increasing data accuracy, strengthening system resilience, and promoting scalability in various industries. These studies offer proof for the notion of decentralized and secure systems that can adjust to the complexities of modern networks, such as the intricate operations of aviation and the large networks of the Internet of Things (IoT).

After conducting a thorough examination of the tabulated data, it is clear that the “Proposed Framework” is a robust and comprehensive strategy for utilizing blockchain technology, specifically in the aviation industry. In comparison to contemporaneous scholarly contributions, this work proposes a

multifaceted technique that encompasses many tiers of system design, including blockchain, data management, access management, networking, and presentation layers. This integrative method distinguishes itself from other research in the review by frequently having a broader reach, encompassing multiple architectural levels, and exhibiting greater specificity.

The “Proposed Framework” showcases its methodological sophistication by utilizing inter-cluster, intra-cluster, and hybrid clustering techniques, distinguishing it from comparative studies that lack in-depth analysis of data segregation and network management. This sophisticated approach demonstrates enhanced effectiveness and capacity in handling more substantial workloads, thus addressing a notable deficiency in research. Moreover, it employs a hybrid storage strategy that integrates on-chain, off-chain, and selectively requested data storage technologies. This advanced method distinguishes itself from the predominantly on-chain storage choices observed in previous research, thereby improving both the accessibility and security of data. This state-of-the-art storage solution surpasses

existing systems by offering a tailored approach that can adjust to various data requirements.

Regarding access management, most of the research surveyed, including our suggested framework, employ Role-Based Access Control (RBAC). Moreover, it improves its functionality by integrating Public Key Infrastructure (PKI) and Access Control Lists (ACLs). This showcases a commitment to developing numerous layers of security protocols, significantly enhancing the system's resistance against unauthorized access. Furthermore, the "Proposed Framework" is notable for its practical validation through the establishment of a test bed, a feature that is not consistently clarified in prior research. This empirical proof demonstrates the effectiveness of the subject in real-world situations, which is sometimes missing in theoretical studies.

The design integrates many techniques for maintaining secrecy, including PKI, data offloading, and ACLs, to emphasize the protection of confidentiality and data privacy. This approach distinguishes itself from and surpasses previous methodologies, such as zero-knowledge proofs, by employing a more diverse strategy to guarantee anonymity.

The paper presents a comprehensive and advanced methodology for leveraging blockchain technology across several businesses, particularly in the aviation sector. The system is notable for its extensive system design, advanced storage solutions, enhanced access control, empirical validation, and robust privacy safeguards. This research provides a substantial contribution to the current body of literature and addresses areas of knowledge that were previously lacking. It demonstrates the system's readiness for usage and its ability to be expanded in real-world scenarios, signifying significant progress in blockchain technology.

4 | PROPOSED FRAMEWORK

This section describes the proposed framework's modular architecture for data management and deploying a blockchain that stores or retrieves desired data while maintaining confidentiality, availability, and integrity.

The primary factors influencing the framework's structure are given in Figure 3:

Contract development, blockchain technology, management of external data requests, and storage systems used are designed as independent plug-and-play modules. This framework, which can be customized according to the problem domain, aims to provide flexibility, and scalability and support the applications of different technologies and protocols.

Data management should be seen as the less invasive usage of immediate and fixed records. Therefore, in storing and updating the data, the node in communication with the base station and the node in between should be separated from each other. In other words, the blockchain database (BCdb) to be created as an inter-cluster (I_c) and BCdb to be created as an intra-cluster (I_a) are separated as characters. As details are given in the following sections I_c stores common data for all participant nodes, while I_a stores data within the cluster. It would be more appropriate to proceed with different concepts since the data in the systems on the nodes are not exposed to exter-

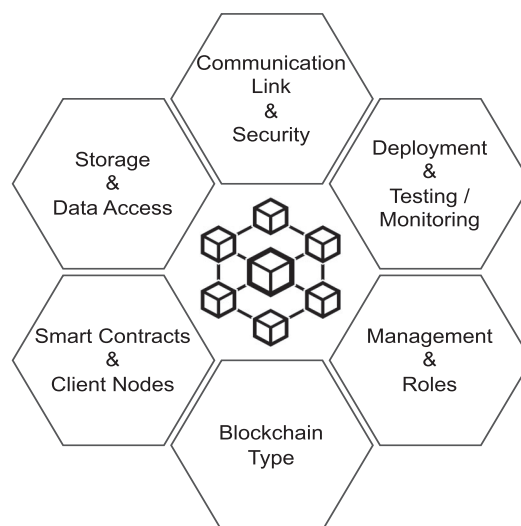


FIGURE 3 Main features that affect the structure of a framework.

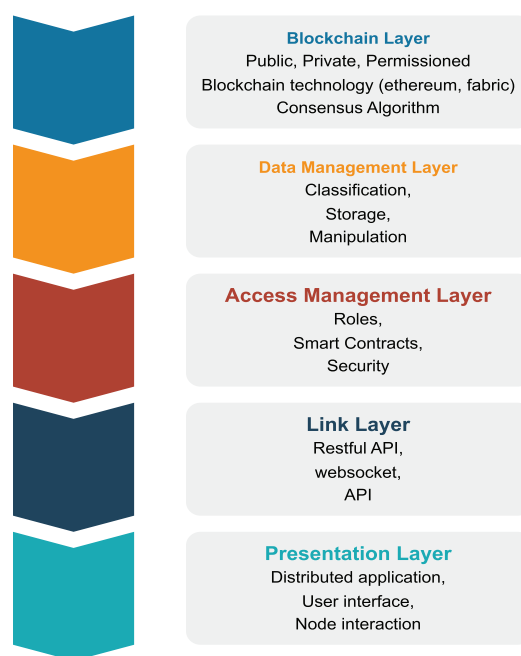


FIGURE 4 Proposed framework layer structure.

nal interference, and the environment that will cause a decrease in communication quality between the base station and the node is always possible. As investigated in literature and given in Section 3, the most prominent demand seems to be the need for a distributed database, management (of data, partners, roles, and identity), sharing, mutual agreement, autonomous or semi nature of self-decision, monitoring and consisting communication link between heterogeneous protocols/devices. A layer-based architecture is built to address the needs as shown in Figure 4.

The layered architecture chosen for the proposed blockchain-based framework aligns with the broader academic and practical consensus on structuring blockchain systems to address scalability, security, privacy, and efficiency challenges in data

management [80]. This comprehensive approach is reflected in the selection of blockchain, data management, access management, link, and presentation layers, each catering to specific functional requirements and challenges inherent in distributed data management systems. The use of clustering to define network segments, hybrid evolutionary algorithms for optimization, and strategic layering for managing data, access, and interactions exemplifies the sophisticated integration of blockchain technology to meet complex system requirements. The selected parameters in the suggested framework are crucial for establishing a strong, adaptable, and safe blockchain-based data management system. The framework's adaptability to different issue domains is facilitated by key features such as contract development, blockchain technology, handling of external data requests, and storage systems. They enable customization based on unique requirements, ensuring the framework's flexibility and scalability. This method enables the incorporation of many technologies and protocols, allowing for a less intrusive utilization of data with safe and efficient storage and updating processes. The modular design allows for easy adaption and integration, making the framework useful across various applications and contexts.

The blockchain layer is crucial for ensuring the security and trust model of the system by choosing the type of blockchain (public, private, permissioned), consensus processes, and underlying technologies (such as Ethereum, Hyperledger etc.). Choosing between public, private, and permissioned blockchains allows for customized solutions that strike a balance between transparency and control, according to unique domain requirements for accessibility and privacy. Ethereum provides a strong foundation for decentralized applications through its smart contract feature, while Hyperledger is particularly effective in controlled enterprise settings that prioritize privacy and efficiency [81]. Consensus algorithms like Proof of Work (PoW), Proof of Stake (PoS), and Delegated Proof of Stake (DPoS) are essential for ensuring the security, integrity, and performance of the blockchain. Choosing the right algorithm is critical for the system's efficiency and trustworthiness. The data management layer is crucial for handling data by organizing, storing, and processing it, while also tackling scalability and efficiency issues. The system can establish a balance between immutability and scalability in large-scale deployments by categorizing data and utilizing several storage methods, both on-chain and off-chain, to address substantial data quantities [82]. Evolutionary clustering in IoT environments demonstrates the use of methods to enhance data management and network performance. Emphasizing roles and smart contracts in the access management layer guarantees that access control is secure and can easily adjust to evolving needs. This corresponds to the need for detailed access controls in settings such as healthcare and IoT, where user responsibilities and permissions need to be carefully handled to protect sensitive data. The link layer tackles the practical difficulties of network interoperability and connection in dispersed systems by facilitating communication across heterogeneous nodes. By incorporating technologies like RESTful APIs and WebSockets, the blockchain framework can effectively communicate with other systems and devices, which is

essential for IoT applications requiring efficient communication across heterogeneous devices. The presentation layer acts as a bridge between the blockchain system and users, guaranteeing that the advantages of the blockchain can be easily accessed by end-users through user-friendly interfaces and distributed apps. User engagement with the system is crucial in fields such as healthcare, as it can greatly influence its effectiveness.

To give a description of the proposed framework's architecture that is illustrated in Figure 5, first cluster, node, and role structure are described. Then the managed data concept is described deeply. Afterward, contracts of design and security concerns are elaborated.

4.1 | Cluster, node and role structure

The blockchain network is designed to support both inter-cluster (I_eC) and intra-cluster (I_aC) operations, increasing its versatility. The I_eC serves as the main data storage and performs as a central communication center, providing connectivity among all participants. This framework differs from the intra-cluster model, I_aC , which functions based on its unique rules and structures designed to fulfill its special needs.

The I_eC plays a crucial role in facilitating communication among distinct I_aC s, effectively connecting multiple clusters. This division into inter and intra-clusters allows for the utilization of different blockchain technology and consensus algorithms tailored to the specific requirements of each cluster. The variety of blockchain technology, each with distinct capabilities, limits, and strengths, requires a flexible strategy. This enables the selection of blockchain solutions that best match the individual objectives of each cluster, assisted by the I_eC and I_aC framework.

Furthermore, the I_eC is crucial in reducing possible delays in data transfer and database growth resulting from significant data exchanges inside the I_aC network. The integrity and immutability of data throughout the network are ensured by the I_eC through storing general-purpose data and retaining hash values of processed data within I_aC s.

The I_aC is designed to be flexible, allowing for the utilization of different consensus algorithms and blockchain technology depending on structural requirements. Engaging with or becoming a part of the I_eC necessitates following the stated norms and protocols of the I_eC . This architecture guarantees the operational autonomy of I_aC s and maintains the network's integrity by keeping data proofs controlled within I_aC s in the I_eC . This strategy highlights the dedication to preserving data accuracy throughout the clusters, demonstrating the framework's capacity to meet various technological requirements while retaining unified network management.

4.1.1 | Inter-cluster (I_eC) architecture

Common data for all nodes and immutability proof (hash) of the I_aC data are determined by the I_eC blockchain. With the help of the architecture, I_eC takes part as a communication bus between

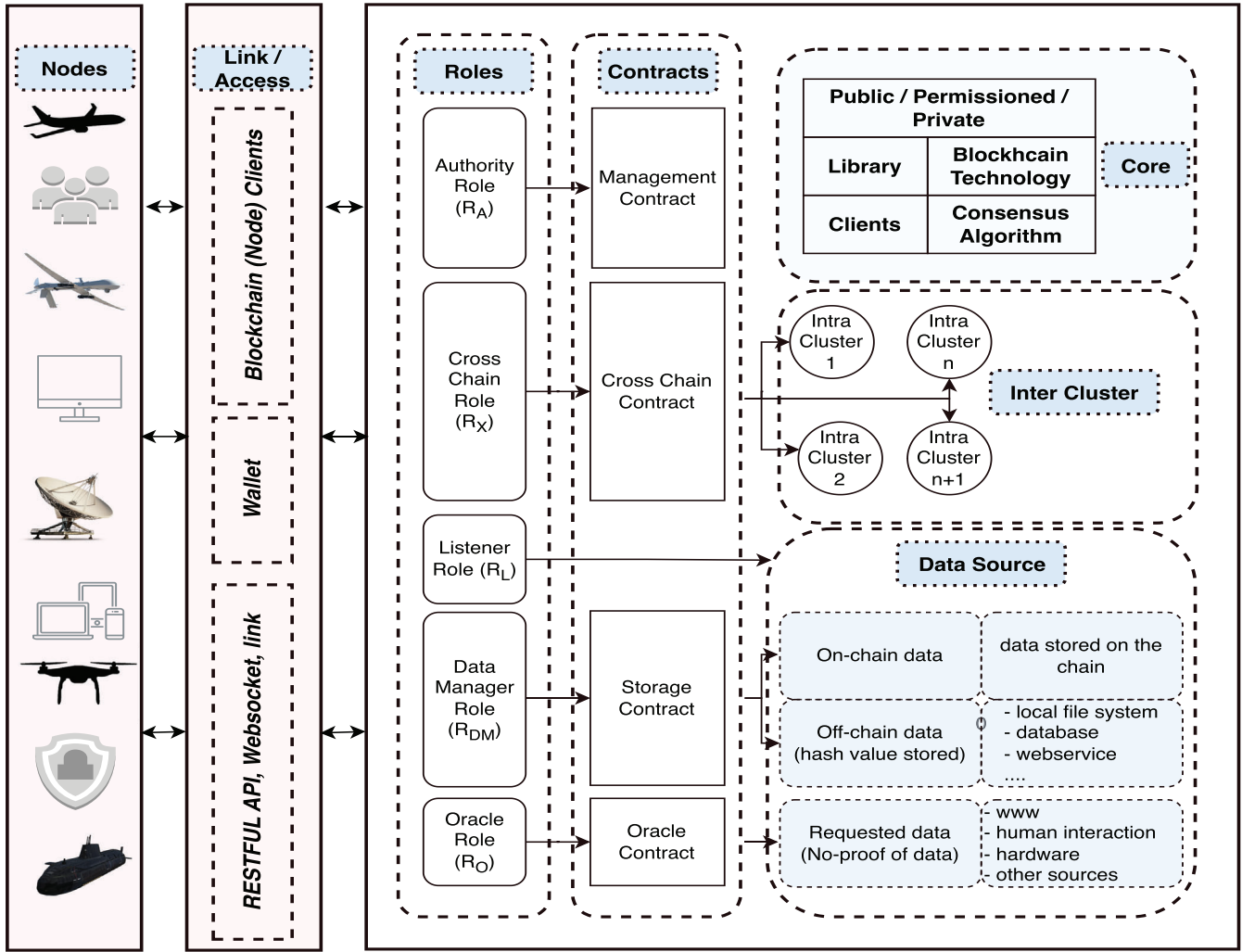


FIGURE 5 Modular architecture of the proposed framework.

I_aC s as well as being the main chain for the framework. Common and general-purpose data which is requested by all nodes is stored and updated on the I_eC chain. Common Operational Picture (COP) and Recognized Air Picture (RAP) creation is considered to illustrate the stored data in I_eC . Aircraft, UAVs, IoTs, or other nodes including ground stations and base stations are responsible for providing navigational information to provide a COP or RAP. Moreover, any node that participates in the blockchain and operates in branches may share appropriate navigational information with the aircraft or any other node in the communication range. At this point, a portion of the data that is gathered by responsible I_aC is passed to I_eC , and aggregating the data in the I_eC maintains the overall picture of COP or RAP.

4.1.2 | Intra-cluster (I_aC) architecture

Prioritizing flexibility and specificity is crucial to address diverse operating needs. Each I_aC in the system has the freedom

to customize its consensus process to align with its own objectives, enhancing effectiveness and adaptability. The I_aC structure empowers customers to select the most appropriate blockchain technology for their unique requirements, facilitating a tailored approach to data management and security. I_aC is a specialized cluster capable of establishing its technology and regulations in a framework apart from I_eC . Although I_aC governs its cluster independently from I_eC , proof of data processed in I_aC can be kept in I_eC to maintain data integrity.

I_aC chain sets its own rules and is formed by nodes with the same level of connectivity such as a swarm environment. I_aC nodes and cluster rules can vary from the I_eC chain they belong to in terms of the consensus algorithm and blockchain technology as well as other used technologies. In this way, cluster formation can be designed in a more flexible and modular manner. To illustrate; while military UAVs flying in swarms and ground support units can form an I_aC with faster and trusted nodes using a few validators or signers, another I_aC chain that serves in an open and hostile environment can employ

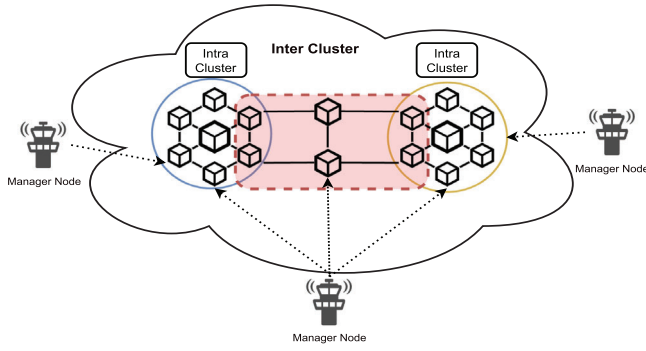


FIGURE 6 Communication between I_aC s through I_eC .

fully Byzantine Fault Tolerance (BFT) consensus algorithms to increase trust even transaction throughput decreases.

4.1.3 | Communication between clusters

Each I_aC can be constructed utilizing a variety of technologies. We establish connectivity between clusters using web sockets and RESTful APIs in a microservice architecture. This architectural option enables effective, scalable, and adaptable communication among clusters. An I_aC node can participate in I_eC by using a connection adaptor that is tailored for I_eC . Any node in a I_aC can communicate with nodes in other clusters, but usually, only one node per cluster is chosen to manage this inter-cluster communication for reasons like energy efficiency or reducing complexity, particularly in applications such as swarm flights.

Each I_aC has at least one node that is responsible for ensuring connectivity with the I_eC . Otherwise, it creates its inter-cluster structure and can not have global proof of the data. Data request or push can be done in three ways:

- From inter-cluster to intra-cluster ($I_eC \rightarrow I_aC$): This method entails transferring data from the inter-cluster network to a designated intra-cluster, enabling external data to be inputted into a cluster.
- From intra-cluster to inter-cluster ($I_aC \rightarrow I_eC$): This entails transmitting data from an intra-cluster to the inter-cluster network, enabling a cluster to share its data with other clusters.
- Between intra-clusters through inter-cluster ($I_aC \rightarrow I_eC \rightarrow I_aC$): This technology enables data transmission between two intra-clusters via the inter-cluster network, enhancing communication between distinct clusters.

The data requested from the I_eC and the commands transmitted by the authorities involved in the I_eC are executed on the nodes that have adaptors of I_eC . At the same time, the data propagated from the I_eC is propagated in the I_aC by necessity or blockchain rule. Therefore, communication between different I_aC s is done via the I_eC chain as illustrated in Figure 6.

TABLE 2 I_aC and I_eC node roles.

	I_aC	I_aC and I_eC	I_eC
Authority role (R_A)	✓	✓	✓
Data manager role (R_{DM})	✓	✓	✓
Listener role (R_L)	✓	✓	✓
Oracle role (R_O)	✓	✓	✓
Cross-chain role (R_X)	X	✓	✓

4.1.4 | Node and role architecture

Nodes are divided into two as their involvement: Inter-cluster node (I_eC) and intra-cluster node (I_aC). Each node can have several roles: authority role (R_A), data manager role (R_{DM}), listener role (R_L), cross-chain role (R_X), and Oracle role (R_O). A summary of the roles that can be held by nodes is given in Table 2.

Authority role (R_A): Responsible for managing the data, communication, and contracts both in inter or intra-cluster domains.

Data manager role (R_{DM}): Responsible for storing and updating data as well as managing access to requested data.

Listener role (R_L): Solely listens to the blockchain for changes. In this role, a node does not have any privilege to update the state of the blockchain.

Cross-chain role (R_X): Responsible for preserving communication links among intra and inter-cluster chains.

Oracle role (R_O): Responsible for populating the requested untrusted off-chain data to nodes. Subscribes to Oracle contract, gets requested data and responses back. Can either be in an intra-cluster or inter-cluster chain.

Inter-cluster node (I_eC): A node is a member of I_eC chain can have one or more of the listed roles: cross-chain (I_eC -X), authority (I_eC -A), data manager (I_eC -DM), Oracle (I_eC -O) and listener (I_eC -L).

Intra-cluster node (I_aC): Communication between custom local chain except I_eC nodes takes part in here. Permissioned and private also customizable blockchain for purpose can be deployed. Crosschain (I_aC -X), oracle (I_aC -O), authority (I_aC -A), data manager (I_aC -DM) ve listener (I_aC -L) roles can be obtained by the I_aC nodes.

4.2 | Data management

Data under consideration vary in terms of frequency of change, size, source and authenticity. Although storing data on a blockchain guarantees integrity, availability, and immutability, it is expensive as it is replicated across all participants. Therefore, it is necessary to classify the data, especially in the domains which have generally constrained resources. Also, by classifying the data, it is possible to determine the storage space and data

updates. These aspects are described in the approach presented under the classification, storage, and manipulation subsections.

4.2.1 | Classification

The proposed framework categorizes data as on-chain, off-chain, or requested based on characteristics such as data type, security requirements, and storage demands. On-chain data is consistently accessible and unchangeable on the blockchain, making it suitable for storing important or confidential information that can benefit from decentralized transparency and resistance to tampering. Off-chain data is information that stores only the proof (hash) on the blockchain, utilizing external storage systems like the Inter Planetary File System (IPFS) for scalability and flexibility, while ensuring integrity through blockchain validation. This method is appropriate for larger files or additional information that does not need on-chain visibility. The requested data consists of real-time information obtained from external sources and is not kept on the blockchain. This data's integrity and quality are maintained using an Oracle architecture that consolidates and verifies data from credible sources, meeting the requirement for dependable external data like meteorological information or wind speeds.

This strategy enables the framework to utilize the advantages of each category, including the security and transparency of on-chain data, the scalability and flexibility of off-chain data, and the dynamic nature of requested data.

On-chain data: It is the data that is always available and reachable, also, stored and updated always directly on the chain. Data within the proposed blockchain framework is stored on the blockchain through a specific storage contract. This contract stores data and regulates access according to preset roles, allowing only authorized individuals to access or alter the information. Access control is essential for preserving the integrity and secrecy of on-chain data, which is secure, transparent, and unchangeable because of its blockchain storage. Data can be queried by the participants whenever needed. This type of data is decentralized and unchangeable, which means it cannot be changed once put into the blockchain. This signifies that the data cannot be directly updated. Instead, updates are handled by the smart contract by creating new transactions that reflect the necessary modifications. These transactions are added to the blockchain, resulting in an auditable and chronological record of data modifications. On-chain data can only be stored and updated by the authorized nodes, in this context, participants who have R_{DM} . On-chain data is frequently utilized for generally crucial or sensitive information. Storing this data on the blockchain ensures transparency, trust, and tamper resistance.

Off-chain data: Data that is not stored on the chain except for the proof of the data (hash) which is stored on the chain for consistency requirements targeting immutability and integrity. Moreover, off-chain data is also referred to as trusted like on-chain data, since integrity proof of the data on the blockchain can be validated with a simple hash validation function. While data is stored in Inter Planetary File System (IPFS) [83], database management system (DBMS), file servers, or in vari-

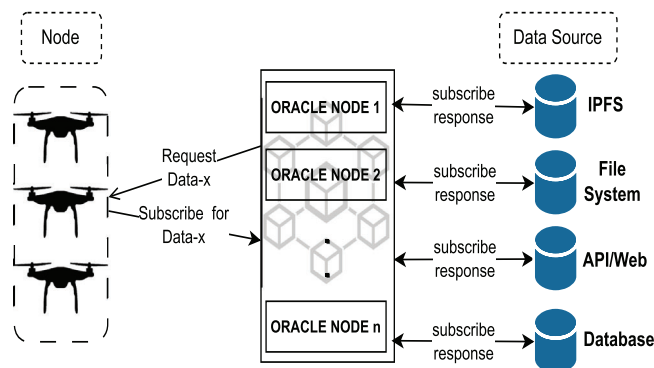


FIGURE 7 Requested data aggregation by Oracle nodes.

ous mediums; the reliability of the data is ensured by storing the proof of the data on the chain. Off-chain data can include larger files, metadata, or additional information that supplements on-chain data. Off-chain data is used to decrease the blockchain's storage and processing constraints, as well as to accommodate data that does not require the same level of transparency or immutability. The data evidence, a hash, and the link to the data are kept on the blockchain within the storage contract in this category. The files or information are saved in external or offline storage systems like traditional databases.

Requested data: Requested data is information that is dynamically acquired or received from outside sources in response to specific searches or user requests. Neither the data itself nor proof of the data are stored on the chain but are fetched as needed. The key difference here is the integrity, accuracy, and validity of the data is in doubt. Since the requested data is in doubt, Oracle architecture is integrated into the proposed framework to make nondeterministic requested data usable in a deterministic blockchain. The requested data is gathered by the Oracle nodes from the appropriate sources and is voted with the aid of a smart contract to determine which data is the most correct as seen in Figure 7. The requested data is dynamically retrieved from external sources and is not stored directly on the blockchain. Events concerning the retrieval of this data are broadcasted on the blockchain, and the details can be retrieved from query logs.

4.2.2 | Storage

Data can be saved in blockchain systems in the form of logs, events, and contracts. Logs are a necessary component of blockchain systems that are built as a linear sequence of entries, each of which represents a specific event or activity. They provide a transparent record of network activity, allowing members to audit and verify the chronological order of occurrences. Logs are normally accessible to all blockchain network participants. The log entries can be retrieved and analyzed by anybody, allowing for transparency and accountability. Events are a higher-level abstraction built on top of logs that enables smart contracts in blockchain systems to communicate with one another and notify interested parties about specific occurrences.

Each event has a name and can have a different set of indexed or non-indexed data than logs, allowing for greater flexibility in capturing useful information. They enable individuals to subscribe to specific events and receive notifications when those occurrences occur. When a smart contract emits an event, a matching log entry is created and recorded on the blockchain. Contract storage is the persistent storage space provided by a smart contract in a blockchain system. Contract storage is similar to a key-value store within a smart contract. It enables the definition and management of variables or data structures that store persistent state information. Contract storage is used to keep and update the smart contract's state. It allows for the storage of data that must be retained over several invocations or transactions.

Data is available in three forms in line with the data classification: on-chain storage, off-chain storage, and requested storage (temporary).

On-chain storage: On-chain data refers to always available and accessible data. This data is stored and updated by $I_{eCN-DMR}$ who owns R_{DM} . Data can be kept on-chain in logs, events, and contract storage. Logs are a necessary component of blockchain systems that are built as a linear sequence of entries, each of which represents a specific event or activity. Even querying logs for specific information is expensive in the context of processing power, the log entries can be retrieved and analyzed by anybody, allowing for transparency and accountability. Events are a higher-level abstraction built on top of logs that enables smart contracts in blockchain systems to communicate with one another and notify interested parties about specific occurrences. Querying is more effective than logs because each event has a name and can have a different set of indexed or non-indexed data, allowing for greater flexibility in capturing useful information. On the other hand, contract storage is the persistent storage space provided by a smart contract which is used to store and update the smart contract's state. It enables the definition and management of variables or data structures that store persistent state information. Data storage in contracts is more expensive than storing in logs or events. When confidentiality is not a concern, storing data in logs is a more effective approach with the ability to index log event topics even when it is limited. However, because it is accessible to all nodes that can access the blockchain, this method is not always desirable. According to the circumstances, either strategy may be applied in the approach that is being provided.

Off-chain storage: Blockchain technologies recognize that not all data needs to be stored directly on the chain. Decentralized file systems or distributed databases can be integrated with blockchain to handle larger or less frequently accessed data. These solutions store the data separately while utilizing the blockchain for validation and verification purposes. Similar to on-chain storage, this storage differs in that it just stores the proof (hash value) of the data and access link information rather than the entire data set. The proof of the data is stored either within the logs, events, or contract. In other words, the proof is kept on the chain but the data itself is kept off the chain. This data is either highly sensitive, large, or utilized so infrequently

that it is not worth regularly adding to the chain's data size. On the chain,

Requested storage: Other data that is not on the chain and is held off-chain but does not have a hash or access information on the chain is referred to as requested data. This type of data is fundamentally distinct from on-chain and off-chain data since it is actively obtained from other sources and not saved on the blockchain as details explained in Section 4.2.1. The Oracle architecture ensures the integrity and trustworthiness of requested data by utilizing a consensus method among trustworthy nodes to validate and aggregate the most correct data. This solution guarantees the integrity of requested data by eliminating the necessity for direct hash storage on the blockchain, therefore resolving issues related to data tampering and alteration. An Oracle contract allows access to and dissemination of this data, which may be queried and accessed in any predefined format by authorized nodes.

4.2.3 | Manipulation

The decentralized and transparent nature of blockchain technology makes it difficult for attackers to change data without being caught. Because of the immutability of past data and the consensus procedures used by blockchains, blockchain systems are more resistant to unauthorized data manipulation attempts than traditional centralized databases. Authorization is the key feature of the proposed framework to ensure the accuracy and integrity of the data.

Roles are defined for several purposes as R_{DM} is responsible for the management of data manipulation by authorized nodes. The node authorized to manipulate (insert, update, and delete) the data that is stored on-chain or off-chain, submits the necessary information via the developed contract and ensures that the data is stored as it is intended.

Nodes in the cluster can access on-chain data that has been updated by R_{DM} as needed. When data is altered, the network propagates the change information through events to reduce the network workload caused by nodes that are querying the needed data for tracking change. As a result, listener nodes monitor propagated data change events in order to update locally used data or maintain local databases up-to-date. Nodes may be asked to confirm that they received the data change event in particular instances. If R_{DM} requests a data change event confirmation, nodes return the response via contracts.

Off-chain proof of data modification follows the same procedure as on-chain data. Off-chain data copies can be kept in many storage devices to balance the access burden. When off-chain data is kept on different mediums and the proof is modified, the blockchain broadcasts a data proof change event to ensure that all copies of the data are changed.

4.3 | Contracts

Smart contracts are computer programs that allow two parties to reach an agreement without the involvement of a

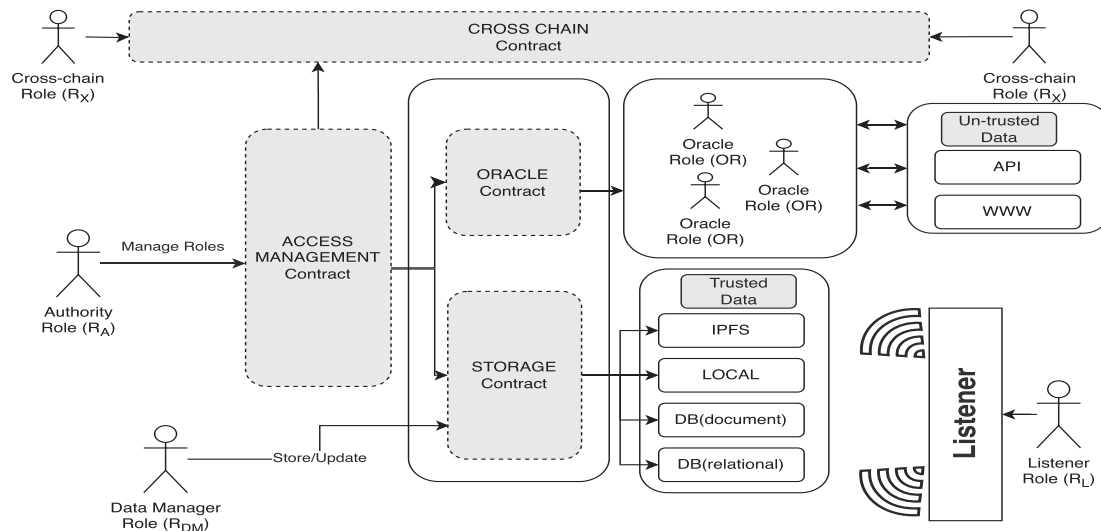


FIGURE 8 Proposed framework contract interaction according to defined roles.

third party. In this method, agreements may be struck without the requirement of a trustworthy common point, particularly a single authority, and the agreed events may be automatically activated when specific predetermined circumstances take place.

As described in more depth in the following sections, the contracts created have made it feasible to offer access management, data management, and communication connectivity. Interaction between contracts and roles are illustrated in Figure 8 and detailed as follows.

Management contract: This contract is responsible for governing and controlling various aspects of the system's operation, typically handling administrative functionalities and defining the rules and policies for the overall blockchain network. The responsibilities, access restrictions, and authorizations that any participating node in the blockchain can have, independent of their position, are defined by the management contract. This contract makes it possible to specify and approve the nodes that are permitted to access storage in accordance with the data classification. Management contracts act as the central authority for decision-making, system configuration, and protocol upgrade as well as defining the roles that are responsible for several duties.

Storage contract: Storage contracts serve as a repository for crucial data and information that must be retained on-chain at all times. It ensures the data's integrity, accessibility, and immutability, allowing all participants to validate it. Although, depending on the usage needs, the type and structure of the data to be stored here may differ. A contract that has been deployed cannot be changed, albeit in some cases, bug patches or upgrades are required to resolve a found vulnerability. Because the storage contract is built with a proxy contract pattern, it allows for the necessary adjustments or upgrades. The proxy contract pattern allows contract upgrades to take place without the need for a new address. All storage access is subject to predefined rules.

Oracle contract: It is the contract that enables Oracle nodes with roles defined by the management contract to meet incoming data requests. Each Oracle contract request is broadcast to the Oracle nodes. When an Oracle node receives a request, it gathers data from various sources and sends it to the Oracle contract for voting to determine the most correct and accurate answer.

Cross-chain contract: Cross-chain contract provides communication between different I_aC s as well as communication between I_aC and I_eC . The address and identity of the users defined in this contract are saved for each cluster in which they participate. As a result, nodes with the R_X are also nodes in the clusters to which they belong. They can perform operations and transfer data based on their roles in each cluster.

4.4 | Security

The security section of the proposed architecture explains the use of the STRIDE threat model, which was created by Microsoft as an analysis tool [84]. This paradigm categorizes security vulnerabilities into six classifications: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service (DoS), and Elevation of Privilege, aiding in their identification within a system. Using STRIDE allows for a detailed and varied evaluation of possible weaknesses, ensuring a full analysis of both the system's structure and its operational performance.

The STRIDE technique is particularly skilled at examining the security status of blockchain-based infrastructures. It allows for a thorough and methodical identification and reduction of possible vulnerabilities in different threat categories. By thoroughly analyzing each STRIDE component within the framework's design and procedural dynamics, one can rigorously evaluate the framework's resilience to typical security threats. This analysis not only emphasizes the security strengths of the framework but also pinpoints potential gaps that may

require additional safeguards to strengthen the system's overall defense measures.

The framework leverages the inherent features of blockchain technology, such as its unchangeable nature, encryption, and distributed agreement processes, to tackle and reduce the security issues outlined by the STRIDE model.

4.4.1 | Spoofing

The framework uses strict authentication mechanisms and role-based access control systems to reduce the hazards of identity spoofing. Verifying the legitimacy and completeness of participants in the blockchain network prevents unauthorized entry and possibly harmful actions. The framework sets out requirements for nodes that want to connect to permissioned private networks and open public networks. It uses mechanisms like virtual private networks (VPNs) for network isolation and specific parameters such as network ID, enode address, and port information to improve network security. Authentication in public and private blockchains relies on public-key infrastructure (PKI) to provide safe transactions and identity verification, allowing only authorized entities to conduct transactions on the network. During a transaction, the sender signs it using their private key $K_p r$, and the receiver verifies it using the sender's public key $K_p u$. Verifying the signed transaction is part of the sender's address. Administrators typically have knowledge of the addresses of nodes in private networks.

4.4.2 | Tampering

The immutable nature of blockchain provides intrinsic protection against unwanted data modifications. Consensus across the network is crucial for preserving data integrity, as each block is cryptographically connected to the one before it, forming an immutable chain. The framework improves data security by utilizing off-chain storage options and Oracle contracts to guarantee the authenticity and reliability of data. Role-based access controls, encryption methods, and strict participation guidelines are used to protect the private blockchain environment, guaranteeing that confidential data stays secure and private.

4.4.3 | Repudiation

The framework uses immutable transaction records to prevent repudiation and ensure transparency and accountability for all parties involved. Digital signatures and PKI play a crucial role in authenticating participant identities, while authorization processes rigorously manage access rights and permissions in the network. The framework prioritizes data confidentiality by utilizing off-chain storage and encryption to protect sensitive information, while also considering the balance between security and resource efficiency.

4.4.4 | Information disclosure

The framework emphasizes data protection through the use of encryption and secure communication protocols to prevent unauthorized access to information. Utilizing methods like transport layer security (TLS), network segmentation, and firewalls is essential for ensuring secure communication and safeguarding sensitive data in a permissioned blockchain setting.

4.4.5 | Denial of service (DoS)

The blockchain's distributed architecture reduces the likelihood of DoS attacks. Decentralization removes individual weak spots, improving the system's ability to withstand challenges and function consistently.

4.4.6 | Elevation of privilege

The framework uses role-based access controls and smart contract functionality to prevent unwanted privilege escalation. Users are only able to perform actions they have been explicitly authorized to do, which helps maintain rigorous control over system operations and access.

The suggested security methodology greatly improves the security of data management systems, especially in industries where data sensitivity and integrity are crucial. Implementing specific security measures within the blockchain structure can effectively combat various threats, offering a strong defense mechanism. However, additional research and practical implementation are required for future endeavors.

4.5 | Integration and challenges

Considering the obstacles mentioned and the proposed framework based on blockchain in the article, the integration process may encounter certain unique issues along with their respective potential solutions:

- **Data compatibility:** The structure of the blockchain framework may not immediately correspond to established data standards. **Solution:** The incorporation of the presentation layer in the suggested framework facilitates the conversion or mapping of current data formats into formats that are compatible with blockchain technology. The link layer facilitates the establishment of connections and enables data access in a predetermined format for existing systems. The data is capable of being modified into various formats and utilized within the system.
- **Interoperability:** Various systems may employ diverse protocols that are not inherently compatible with the blockchain framework. **Solution:** The proposed solution involves creating or using blockchain middleware that can connect with different protocols and systems. This will allow for a smooth

flow of data across the proposed link layer. APIs and web sockets facilitate the interchange of data between existing systems, even if they use different protocols.

- **Data quality:** The unchangeable characteristic of blockchain could hinder the rectification of data mistakes after they have been recorded. **Solution:** Implemented smart contracts that incorporate pre-validation of data prior to its inclusion in the blockchain, while also establishing and managing off-chain data repositories for data that can be modified. Additionally, the proxy pattern can be utilized to apply changes or patches to deployed contracts in order to ensure data integrity during the lifespan of the contracts. The assurance of data quality can be achieved by the utilization of on-chain, off-chain, and requested data architecture.
- **Efficiency and ability to handle increasing workloads:** Public blockchains, in particular, can encounter challenges related to scalability and performance. **Solution:** Choose scalable blockchain solutions or hybrid models that integrate the security of blockchains with the efficiency of conventional databases. The suggested inter and intra-cluster system allows for the utilization of diverse blockchains and consensus algorithms to ensure scalability based on performance requirements. Certain blockchains exhibit superior performance under huge transaction volumes, whereas others do not. Therefore, the choice of the intra-cluster consensus technique and technology can be made independently of the main inter-cluster, based on the specific requirements.
- **Ensuring security and compliance:** Incorporating confidential information into a blockchain system may give rise to compliance concerns. **Solution:** Utilizing private or permissioned blockchains that restrict access and employ encryption to guarantee compliance and safeguard data privacy. The suggested frameworks' access management layer addresses this issue, as illustrated in the relevant sections.
- **Organizational change:** Resistance to the adoption of blockchain technology arises from its inherent complexity. **Solution:** Implement extensive training initiatives and showcase tangible advantages to relevant parties in order to promote more seamless integration.

To tackle these problems, a combination of technological, operational, and strategic methods must be employed. This will ensure that the integration of the blockchain framework enhances existing data management systems and minimizes any potential interruptions. The evaluation section demonstrates the applicability of the proposed framework to the domains indicated before. However, we acknowledge that there is still room for improvement, and further investigations are required. Therefore, this proposed paradigm is an excellent foundation for data management.

Implementing the newest proposed systems or frameworks in real-world scenarios can encounter initial challenges when discussing the necessity of using and incorporating the new system into an old one that is already familiar to everyone. This paper provides an explanation for the necessity of the suggested framework and outlines the specific challenges that need to be addressed in a systematic manner.

Assessment of existing data management systems: This stage entails assessing the existing data management systems already utilized in the unmanned vehicles and aviation sector. The suggested framework recommends the identification of the functionality, data formats, protocols, and security measures of these systems.

Identification of integration points: After evaluating the current systems, the subsequent task is to identify the specific areas where blockchain technology might be incorporated to supplement or improve these systems. The paper includes examples in several domains, such as flight data recording, maintenance tracking, supply chain management, and regulatory compliance.

Data mapping and conversion: This refers to the process of synchronizing the data structures and formats of current systems with those that are compatible with blockchain technology. The essay proposes the creation of techniques or middleware layers to enable smooth data exchange and conversion between the two systems.

Development of blockchain interfaces: During this phase, interfaces or APIs are created and implemented to facilitate communication and compatibility between the blockchain framework and pre-existing data management systems. The emphasis is on compatibility with current communication protocols and data exchange standards.

Integration testing and validation: Thorough testing is carried out to verify that the integrated system operates as planned. The essay emphasizes the significance of testing many aspects such as data exchange, transaction processing, smart contract execution, and data synchronization in order to verify the functionality, performance, and security of the integrated system. Integration with the existing systems is facilitated through the integration of the presentation layer and connection layer.

Deployment and rollout strategy: This entails strategizing the implementation of the consolidated system in a way that reduces interruptions to current activities. It is advisable to use phased deployment tactics, beginning with pilot projects or proof-of-concept initiatives, prior to carrying out full-scale implementation.

Training and change management: Stakeholders are given comprehensive training programs and change management initiatives to acquaint them with the integrated system. The paper highlights the importance of teaching consumers about the advantages, features, and optimal methods related to the blockchain-based data management system.

Monitoring and maintenance: Procedures are implemented to oversee and uphold the continuous performance, dependability, and safety of the integrated system. Continuous monitoring tools, security audits, and proactive maintenance techniques are utilized to immediately identify and resolve concerns.

Compliance and regulatory alignment: This stage guarantees that the integrated system adheres to pertinent aviation regulations, data privacy legislation, and industry standards. Comprehensive evaluations of regulatory mandates are carried out, and essential measures are put in place to ensure

adherence. Integration with the existing systems is facilitated through the integration of the presentation layer and connection layer.

Iterative improvement and optimization: The integrated system is continuously improved and optimized through the analysis of user feedback, performance indicators, and the incorporation of emerging technology. The book proposes the idea of continuously improving the system to better its usefulness, usability, and efficiency as time progresses.

Given the stringent regulations in the aviation industry, the use of blockchain technology must be approached with caution to ensure compliance with existing norms. In addition to the utilization of Internet of Things (IoTs) and Unmanned Aerial Vehicles (UAVs), the aviation sector is one of the most tightly regulated businesses, with rigorous criteria for ensuring safety, maintenance, and operations. When integrating blockchain solutions, it is crucial to ensure that compliance with existing regulations is not compromised. This requires working closely with regulatory agencies. It is necessary to collaborate with regulatory bodies in order to integrate the unique features of blockchain technology, such as its distributed ledger technology and smart contracts, within the regulatory framework while ensuring safety and adherence to operational standards. The study thoroughly examines the adaptability of the suggested framework, emphasizing its potential to conform to current and future legislation. However, the study also acknowledges the need for further research on the practical application and real-life effects of the framework in relation to adhering to legal and regulatory implications.

Furthermore, the integration of the proposed framework, or any innovative system, into existing infrastructures has other challenges, including the need for technical and non-technical user training. To address these challenges, the framework emphasizes the importance of providing training to both technical and non-technical staff. Although the focus of this study does not include training, it is important to build the training program to accommodate persons who lack familiarity with technical issues. The text highlights the significance of blockchain technology in connection to their particular roles, emphasizing its benefits in enhancing data integrity, security, and transparency. Furthermore, the framework provides detailed guidance on how to efficiently employ the blockchain-based system for specific tasks and underscores the significance of adopting optimal security measures. The training program should also incorporate guidance on roles pertaining to compliance and reporting. This instruction elucidates the manner in which blockchain architecture facilitates the fulfillment of regulatory responsibilities and the streamlining of audit procedures.

The architecture of the proposed framework is inherently hierarchical, facilitating its utilization by technical individuals and enabling seamless integration with existing systems through the utilization of technologies such as APIs and WebSockets. This approach not only makes use of current technology in areas like monitoring but also enables the development of user-friendly interfaces at the presentation layer for individuals

TABLE 3 Blockchain tools for development, testing, and monitoring.

Environment	Tools
Integrated dev. env. (IDE)	VsCode, bash terminal, solidity IDE
Development environment	Truffle Suit, Remix, Metamask, HDwallet Node.js, JavaScript (js)
Library	web3.js, ether.js, custom scripts
Blockchain client	HyperLedger Besu, Ganache
Testing tools	HyperLedger Caliper, Truffle Suit
Monitoring tools	Prometheus, Grafana, web browser

who are not technically inclined. This allows both non-technical users to use the proposed framework independently or in conjunction with other systems.

5 | EXPERIMENTAL EVALUATION

In order to investigate the validation of the presented framework, tests were carried out with an experimental approach. Applications and technologies developed for the realization of this scenario and a summary of them are given in Table 3.

5.1 | Scenario

The Flight Management System (FMS) is a dual system composed of the Flight Management Computer (FMC), Common Display Unit CDU, and a cross-talk bus. Comparison studies present that there are variations in pilot training, aircraft performance, and errors in data collection and processing according to developed FMS [85, 86]. Another problem is the capacity of the FMS which has limited storage to maintain the required data to FMS [87]. Using the FMS database populated from many sources, a common use of pilot entry can be considered to be met. As a result, the NavDB and PerfDB consist of the combination of various data that all the other modules are fed with. While the performance database is not shared publicly NavDB is common data for nodes in the aviation domain. NavDB is updated regularly by the aviation authorities and is the main source of flight planning. For these reasons, proof of concept evaluation is performed on the presented framework for storing NavDB, which is important in the aviation domain and is updated every 28 days. We obtained the real navigation data which contains files and approximately 1.240.158 bytes of formatted plain text data. Example data format is shown in Table 4.

It is ensured that NavDB is stored on the inter-cluster blockchain (I_cC) so that data can be stored, updated, and deleted on the blockchain. The I_cC nodes are assumed to have provided their addresses and participated in the consisted blockchain network, which is designed to be private, and is structured parallel to security considerations. Each node has its own private-public key pair and an address derived from it. The blockchain also

TABLE 4 Navigation data formation.

NAV_IDENT	TYPE	CTRY	NAME	ICAO	WAC	FREQ	WGS_LAT	WGS_DLAT	WGS_LONG	WGS_DLONG	SLAVED_VAR	MAG_VAR	ELEV
ST	5	TU	ATATURK	LT	323	340000K		N40574590	40.962.750	E028481350	28.803.750	E005522 0222	U
BCN	4	SP	BARCELONA	LE	319	116700M		N41182560	41.307.111	E002062810	2.107.806	E001320 0222	0
BML	9	US	BERLIN	KZ	263			N44380072	44.633.533	W071111029	-71.186.192	W014459 0222	1730
.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.	.
.	.	.	.	.	.	.	.	.	.	.	.	.	.
ESB	4	TU	ESENBOGA	LT	323	112100M		N40084780	40.146.611	E033004490	33.012.472	E006056 0222	3150

TABLE 5 Developed programs and used technologies.

Data Type	Program	Description
On-chain	websocket	Program to access data on contract storage
Off-chain	IPFS	Distributed data storage
	CouchDB	Document based database management system
	PostgreSQL	Relational database management system
Requested	API	event handler for requested API data with parameters
	web listener	js-based program for search data on www

*Used libraries: ipfs-http-client, body-parser, request, node-fetch, express, axios, dotenv

includes validator nodes with predefined nodes that are granted with R_A . The R_A node owns the right of the management of private blockchain and has deployed the management, storage, and oracle contracts, respectively. The NavDB data, which contained real data, was formatted and added to the storage contract by the node having the R_{DM} before being updated. The R_L nodes obtained the requested data by sending a read request to the NavDB data stored on the storage contract or by listening to the events that emitted from the chain.

5.2 | Development environment

The Ethereum blockchain network is used because it enables smart contracts, a well-known and preferred technology in the literature, and it is available to the public. The proposed framework is constructed as a private network. VsCode IDE, the Truffle suit extension, and Remix Ide are used for developing smart contracts in Solidity language. Smart contracts are tested on the Ganache Ethereum network before being deployed into production.

To meet the security needs of the production environment, the HyperLedger Besu client is utilized to establish the private blockchain. To test a single transaction to request or gather data, HDwallet and Metamask are used to sign and send transactions (txs). Node.js, web3.js, and ethers.js libraries are used to interact with the blockchain via web socket and API. Applications are developed using the Javascript language to initialize the proposed framework with its components and evaluate tests. Web socket, API, and web event handler applications are developed to realize access to on-chain, off-chain, and requested data. Applications' responsibilities include serving the on-chain data as requested, listening to logs and events, gathering off-chain data from its address and checking its hash value for consistency, performing Oracle voting, and gathering it from different sources for requested data. With these applications, access to data in different structures such as IPFS, CouchDB [88], and PostgreSQL [89] is made possible. An auto-run application is developed to initialize the proposed blockchain framework and test it with flexible parameters. The developed applications, the used technologies, and their brief descriptions are given in Table 5.

```

{
  "config": {
    "chainid": 2929,
    "berlinBlock": 0,
    "qbft": {
      "epochlength": 30000,
      "blockperiodseconds": 2,
      "requesttimeoutseconds": 4
    }
  },
  "nonce": "0x0",
  "timestamp": "0x58ee40ba",
  "extraData": "0xf87aa00...",
  "gasLimit": "0x47b760",
  "difficulty": "0x1",
}

```

FIGURE 9 QBFT genesis file parameters.

5.3 | Test environment

The genesis block files of the IBFT 2.0 and QBFT consensus protocols are created in order to implement the proposed framework. PKI key pairs that are uniquely utilized by each node are generated and applications are developed to provide the necessary data communication among the blockchain network and the nodes. Private key K_{PR} and public key K_{PU} are generated for each node with the help of the 'besu operator batch'. Node.js 18, CouchDB, PostgreSQL, IPFS local clients, and the mentioned libraries are installed on the same workstation which is powered by Intel Core X-series processors, DDR4 4266 MHz 64 GB RAM.

Hyperledger Besu client parameters (chainId, milestone block, protocol, fixed difficulty, block period, epoch length, request timeout etc.) are defined in the genesis file according to the consensus protocol. IBFT 2.0 and QBFT initial genesis file parameters are given in Figure 9.

Since there is no generally accepted test benchmark in the literature, HyperLedger Caliper is preferred due to its applicability and the fact that it provides fundamental values such as throughput and latency metrics which are considered important for validating the framework.

Management, cross-chain, storage, and oracle contracts. aUsers are created and roles are granted for testing purposes. The implemented framework is tested for usability. Storage contract NavDB initialization function is considered for performance testing of the proposed framework. The initialization function algorithm is given in Algorithm 1.

To perform the specified tests, it is necessary to run several iterations in an automated environment. Obtained results must be recorded in an appropriate format in order to conduct a fair comparison. In addition, the same test scenario must be run at least three times so as to obtain a mean average for reliable data analysis regardless of the environment. To address the mentioned requirements, a novel automated testing application is developed in JavaScript language to test multiple scenarios as

parameters are illustrated in Table 6. Each test is conducted on a newly created blockchain. Additionally, contract deployment and granting roles are done in an automated manner. To the best of our knowledge, there is no automated application developed for these types of requirements.

The main goal behind our experimental evaluation is to ensure that the developed framework is fit-for-purpose and fit-for-use, as well as to assess its suitability for the scenario presented. To that end, the proposed framework is tested on an active Ethereum network using the block creation time per second (bps) operator in the genesis file, the number of validator nodes, and a comparison of the IBFT 2.0 and QBFT consensus algorithms. On the other hand, Hyperledger Caliper benchmarking configurations and rate controllers are compared in the context of error rates. The effect of the number of workers (w) and transactions sent (tx) are also examined. Furthermore, five (5) different rate controller approaches are tested: fixed rate, fixed feedback rate, fixed load, maximum rate, and linear rate. Through the developed testing application, the necessary configurations for the HyperLedger Caliper to connect to the running Ethereum network are made, the reports are recorded for each case, and system resources are monitored for further investigation. Our obtained test results and discussion are provided in the following subsection.

6 | DISCUSSION

PoA consensus protocols IBFT 2.0 and QBFT are tested sequentially on the utilized blockchain with the storage contract's initialized function. Storage contract has initialize function (F_{init}), update function (F_{update}), read function (F_{read}), and variations of them. The F_{init} is investigated to validate the proposed framework because it changes the state of the blockchain and is expected to consume more resources according to other functions.

The proposed framework is utilized with IBFT 2.0 consensus algorithm, 4 validator nodes, and 2 block period seconds (bps) parameters. Blockchain is evaluated for number of transaction success with 1, 5, 10, 20, and 50 workers numbers and under 1000 txs load. Also, the effect of different workloads on the transaction success is investigated. HyperLedger Caliper rate controllers are fixed feedback rate (fixed_feedback), fixed load rate (fixed_load), fixed rate (fixed_rate), max rate (max_rate) and linear rate (linear_rate). Obtained results are given in Figure 10. The x-axis represents the number of workers and the y-axis is the success and error number under load. Workload parameter error (_err) and success (_succ) values are represented by adding the end of the workload name such as fixed_feedback_err for fixed feedback rate error. Except for a fixed load rate of up to 10 workers, the number of successful transactions increases in all rate controllers. The fixed load rate controller performs similarly until 10 workers, beyond which the success rate diminishes. Furthermore, when all rate controllers are run with 10 workers, the maximum number of successful transactions is observed, and after that point, the number of

ALGORITHM 1 NavDB *Initialize Function* algorithm.

Data: Valid Parameters including caller identity C , NavDB value identifier N_{id} , and NavDB data N_{data} .

Result: Initializes the NavDB structure with the given data and emits an event upon successful initialization, or returns an error on unauthorized access.

Input: C (Caller identity), N_{id} (NavDB value identifier), N_{data} (NavDB data)

Output: Response data indicating successful initialization or an error message

```

 $C \leftarrow$  Caller;
/* The entity attempting to perform the initialization. */
 $N_{id} \leftarrow$  NavDB value identifier;
/* Unique identifier for the NavDB data entry. */
 $N_{data} \leftarrow$  NavDB data;
/* The actual data to be stored in NavDB. */
if  $C$  can call method then
    create Nav struct object ( $N_{id}$ ,  $N_{data}$ );
    /* Initializes a new NavDB structure with the provided identifier and data. */
    map object navdata[ $N_{id}$ ] = nav;
    /* Associates the NavDB identifier with the new data entry. */
    emit event(NavDB_INITIALIZED,  $N_{id}$ );
    /* Signals the successful initialization of the NavDB entry. */
else
    return error[Unauthorized access];
    /* Ensures security by preventing unauthorized modifications. */
end

```

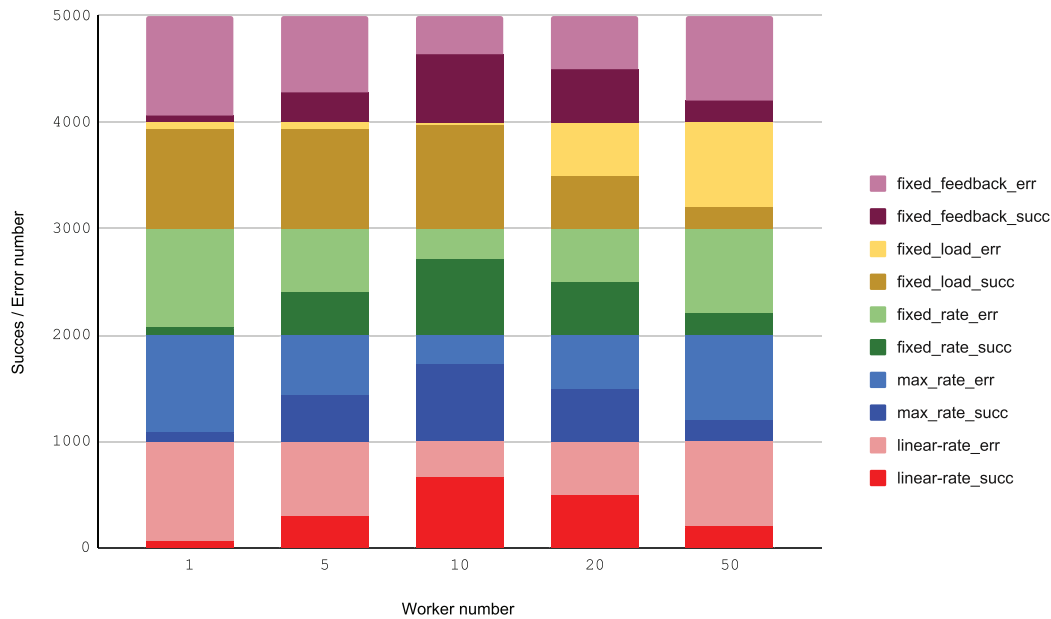


FIGURE 10 Rate controller success/error rates for IBFT 2.0.

successful transactions decreases. As a result, the best number of workers for further testing has been determined to be 10.

The effect of the number of validators is another concern that has a direct impact on block creation time because it increases the time required to validate the blocks to be created. At least, three out of four validator nodes must validate the block in order to respect Byzantine Fault Tolerant (BFT). Figure 11 shows the test results for 50 tx, 2 bps, and 10 workers on an IBFT 2.0 network with 4, 10, 20, 30, and 40 validators. Even

though increasing from 4 to 10 validators resulted in a decrease in throughput and an increase in latency. Increasing the number of validators to 20 resulted in the opposite, however, in a small difference. Throughput is decreased and latency is increased in tests with 30 and 40 validators, respectively. As a result, the expected decrease in throughput with increasing the number of validators in PoA algorithms is interpreted as an increase in latency. The fact that the throughput is higher at 10 validators than at 20 validators can be attributed to network resources

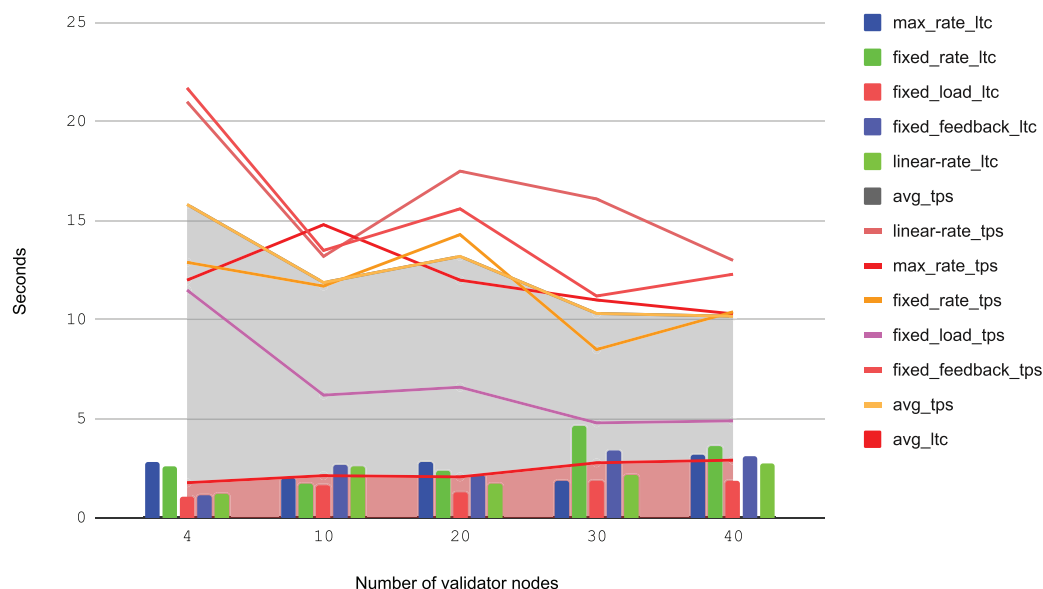


FIGURE 11 Affect of the validator number on throughput and latency.

TABLE 6 Developed automated testing program parameters and definitions.

Parameter	Data Type	Description
folder_name	string	setting name of the top folder
create_folder	bool	option to creation of folder
node_count	int	set validator nodes number
i_q_bft	string	set consensus protocol
change_genesis	bool	set validators and consensus
run_node_count	int	set running nodes count
run_prometheus	bool	run Prometheus data collection
tsdb_name	string	prometheus database name
run_report	bool	run HL Caliper for testing
report_name	string	set generated report name
promet_data_export	bool	export tsdb database
promet_db_delete	bool	delete prometheus database
migrate_change_config	bool	migrate contract, set config

and may be insignificant. The effect of the rate controller on transaction commit time can be seen as average throughput and average latency are represented as *avg_tps* and *avg_ltc*, respectively. To conclude, as the number of validator nodes increases, throughput decreases, and latency increases.

Block period seconds (bps) is the parameter defined in the genesis file to set the creation time of a block even if there is no submitted transaction. QBFT algorithm is investigated for bps 1 and 2 with 4 and 20 validator nodes to examine the correlation between them. Throughput is the highest for all transaction numbers in the test results with 4 validator nodes, 10 workers,

and 1 bps, according to the results with the QBFT algorithm given in Figure 12. Then tests are conducted with 20 validator nodes, 10 workers, and 1 bps. The results of the tests with 4 validator nodes, 10 workers, and 2 bps show that the time it takes to create a block has a direct impact on throughput. Comparing the results validator increase has less effect than bps under 4 and 20 validator nodes. Throughput increases as bps decreases even with more validator nodes. *4n_10w_bps1* (4 nodes 10 workers 1 block period second) gives best throughput, *20n_10w_bps1* and *4n_10w_bps2* follows. At the same time, the latency in all configurations increases as the number of tx increases.

In order to assess the scalability of the proposed system, a comparative analysis of consensus techniques is performed. For comparison of the QBFT and IBFT 2.0 algorithm on the proposed framework 4 validator nodes, 10 test workers, and 1, 2 bps are taken into consideration. Figure 13 shows that QBFT performs better than IBFT 2.0. But, when comparing IBFT results bps 1 outperforms bps 2 in difference with QBFT.

The framework's scalability was thoroughly evaluated by changing the number of validator nodes and analyzing the effects on transaction throughput and latency. The tests were conducted to determine the framework's capacity to efficiently manage growing workloads without seeing a notable performance decline. The testing showed that the framework achieves a balance between throughput and latency until a specific threshold of validator nodes and workload intensity is reached. The study found that the most effective number of workers for testing was 10, resulting in the highest number of successful transactions across various rate controllers tested.

The framework's ability to handle significant transaction volumes was shown through performance testing in high-demand situations. The experiment assessed the framework using a specific amount of transactions under various configurations of the IBFT 2.0 and QBFT consensus algorithms, emphasizing block creation time and the number of validator nodes.

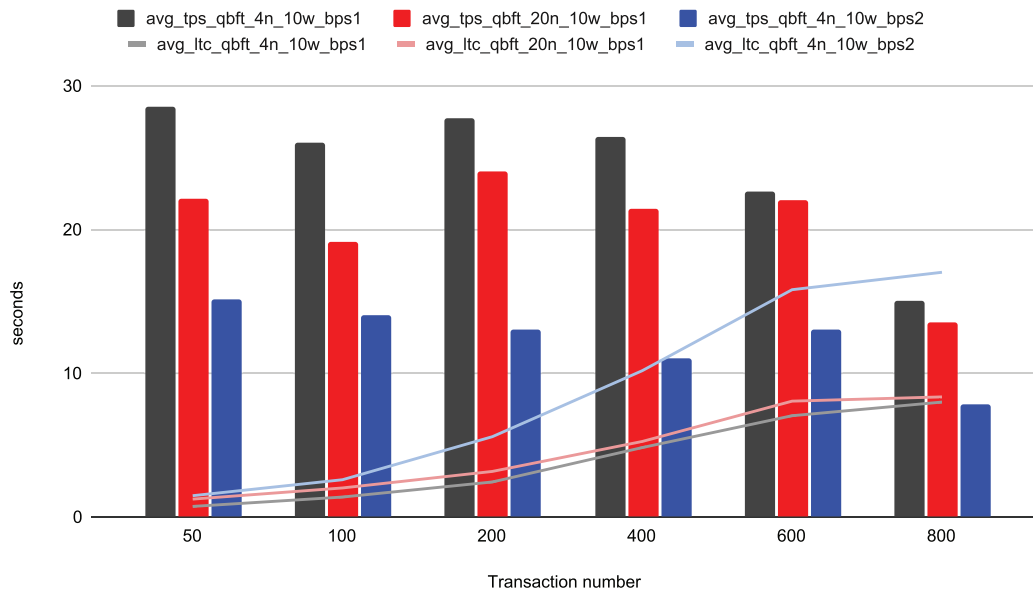


FIGURE 12 Effect of validator node number and bps over throughput and latency on QBFT.

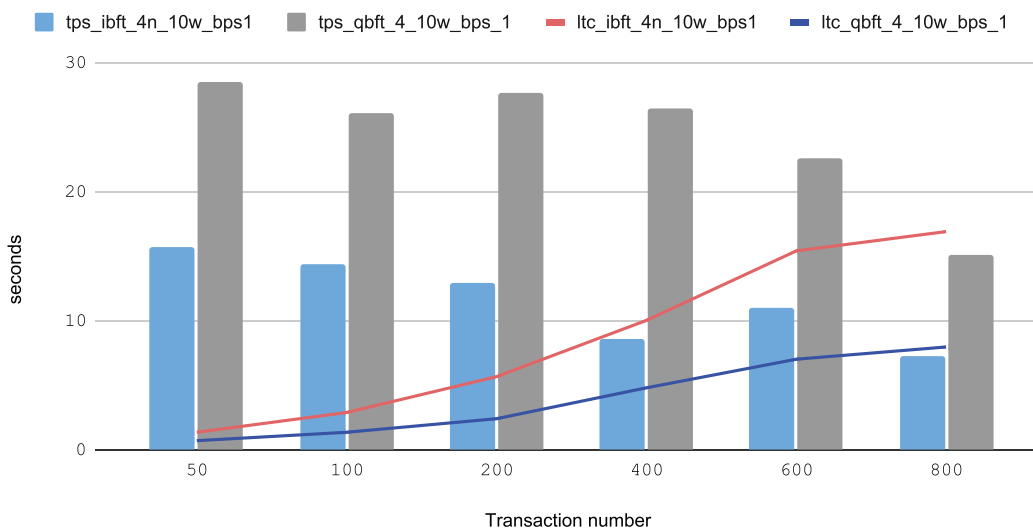


FIGURE 13 Comparison of IBFT 2.0 and QBFT consensus protocols.

The framework's performance is influenced by the consensus mechanism, block generation interval, and the number of validator nodes. A reduction in block period seconds (bps) and an ideal number of validator nodes, discovered through testing, improved transaction throughput while keeping latency levels within acceptable limits.

7 | CONCLUSION AND FUTURE WORK

The goal of the framework is to provide a structure that can be implemented in various domains for data management that are utilized in a clustered or in some conditions in a private environment but communicates with the external parties. As is seen in

the obtained test results and also discussed; with the help of the proposed framework data can be transferred securely, structural decisions can be made autonomously within the intra-cluster blockchain, and the intra-cluster blockchain can interact with another blockchain that manages its own structure apart from any other blockchain through inter-cluster. Inter-cluster enables the communication between intra-clusters and stores common data needed by all participants. The inter-cluster is also responsible for participation in the blockchain, the management of general rules, and the provision of data access, exchange methods, and structures. On the other hand, data integrity has been ensured by introducing access and acquisition methods in accordance with the data structure. Access control and authentication procedures have also been utilized to assure access security. The

technique given has resulted in a framework that is applicable and powerful, particularly in circumstances where there is internal management but also external control or a requirement for communication.

With the developments in blockchain technology and the increasing need for zero defects in the industry, many applications can be developed for managing regulations and solving problems in the context of data management that are encountered in aviation, UAVs, IoTs, smart city applications, product tracing, passport validation etc. is now possible with the help of the framework presented in this study. In the case of the passport control mechanism that uses the presented framework; each country stores its own passport information and this information is stored as private in intra-cluster. The inter-cluster holds the proof of the passport validation information and when all countries are members of this inter-cluster, one country can securely verify the passport information of another country.

There are similar frameworks in the literature, however, all of them are focused on a specific problem or applied to a specific domain. To the best of our knowledge, there is no such framework that can support diverse applications in the literature. In this paper, a baseline framework is presented that can be a remedy to various problems in numerous problem domains in the context of data management.

Experimental evaluation has shown that with the concept validation of the proposed framework, role-based access to data either on or off the blockchain can be provided in accordance with access control utilizing developed contracts and other components. Furthermore, a solution for gathering nondeterministic information from the chain is proposed.

Our research aims to implement and thoroughly test the suggested framework in real-life scenarios, with a particular focus on the Internet of Things (IoT), Unmanned Aerial Vehicles (UAVs), and the aviation industry as future work and open research areas. Our objective is to collect relevant data from real-world implementations by installing the framework onto actual devices and systems and conducting detailed observations. This data will be essential in identifying the necessary enhancements and optimizations, guaranteeing that the framework adequately fulfills the practical requirements of these advanced technical fields.

In addition to the focus on practical implementation, the creation of a dedicated testing environment has also become a secondary goal. While we have made some progress in developing a simulation and test environment to reduce real-world hazards, the many intricacies and contributions of this environment were not within the focus of our current study. Future research initiatives will focus on providing a comprehensive and major contribution to this test environment, developing a strong platform that will greatly help ongoing and future research and development efforts.

The framework's integration capabilities with current real-world systems have been greatly affected by engagements with prominent aviation corporations, namely in the areas of data storage and access procedures. Future endeavors will be focused on implementing the framework with tangible devices within

these companies. This crucial milestone in achieving practicality and verification is anticipated to strengthen the framework's significance and efficiency in real-life settings, thereby enabling a smooth integration with current systems.

Moreover, the ever-evolving domain of consensus algorithms presents a hopeful opportunity to improve the flexibility of the framework. Exploring new domain-specific consensus algorithms has the potential to utilize the inherent flexibility of the proposed framework, incorporating these technological breakthroughs to improve performance and efficiency. Future research will focus on expanding and speeding up these algorithms, to ensure that the framework remains at the forefront of technological progress.

To address the scalability limitations of the framework and assess its maturity, it will be necessary to simulate the framework under high-load situations, employing technologies like load balancers. This strategic approach aims to thoroughly evaluate the framework's performance in different stressful situations, enabling specific enhancements to ensure strong scalability. By conducting systematic testing and assessment, the framework's capacity to endure and function optimally in various settings will be greatly improved.

Finally, ensuring security is of utmost importance, and it is crucial to focus on developing automated and standardized security testing methods based on threat modeling. Our goal is to enhance the security characteristics of the framework by coordinating attacks based on known threat models and closely monitoring the system's overall response. The continuous effort to improve security measures aims to protect the system from constantly changing cybersecurity threats, offering a safe and dependable platform for users in different fields.

Future development includes implementing the system on test Ethereum networks and establishing it as a container, building upon the success of the framework. The objective is to introduce a framework that is simple to implement, practical, and can be expanded to address issues in different fields. Moreover, there is ongoing study in the field of boosting throughput and addressing other security concerns related to hostile actions. The objective is to assess the framework in a setting with many grouped drones powered by Nvidia Jetson Xavier, enabling data interchange during a partially autonomous swarm flight.

AUTHOR CONTRIBUTIONS

Ozan Zorlu: Conceptualization; data curation; formal analysis; investigation; methodology; resources; software; validation; visualization; writing—original draft; writing—review and editing. **Adnan Ozsoy:** Methodology; project administration; supervision; validation; writing—review and editing.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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